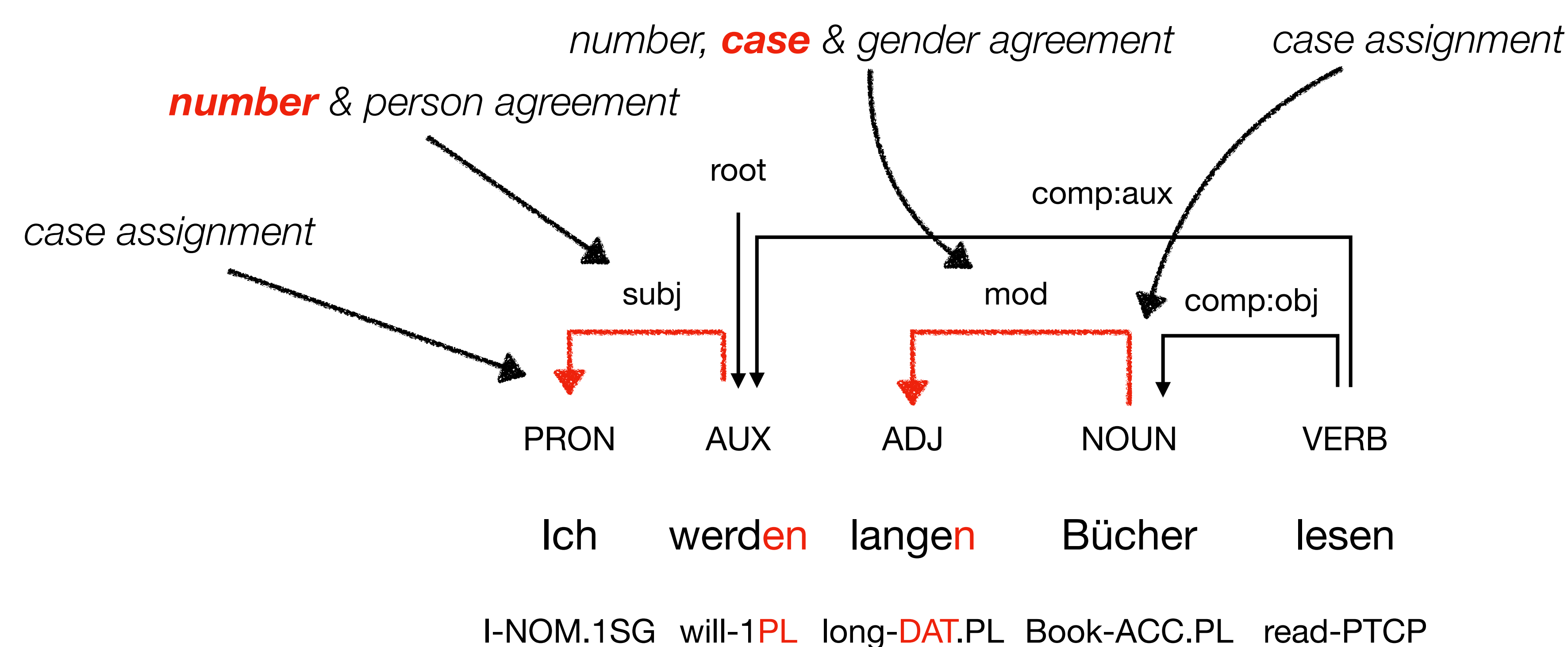
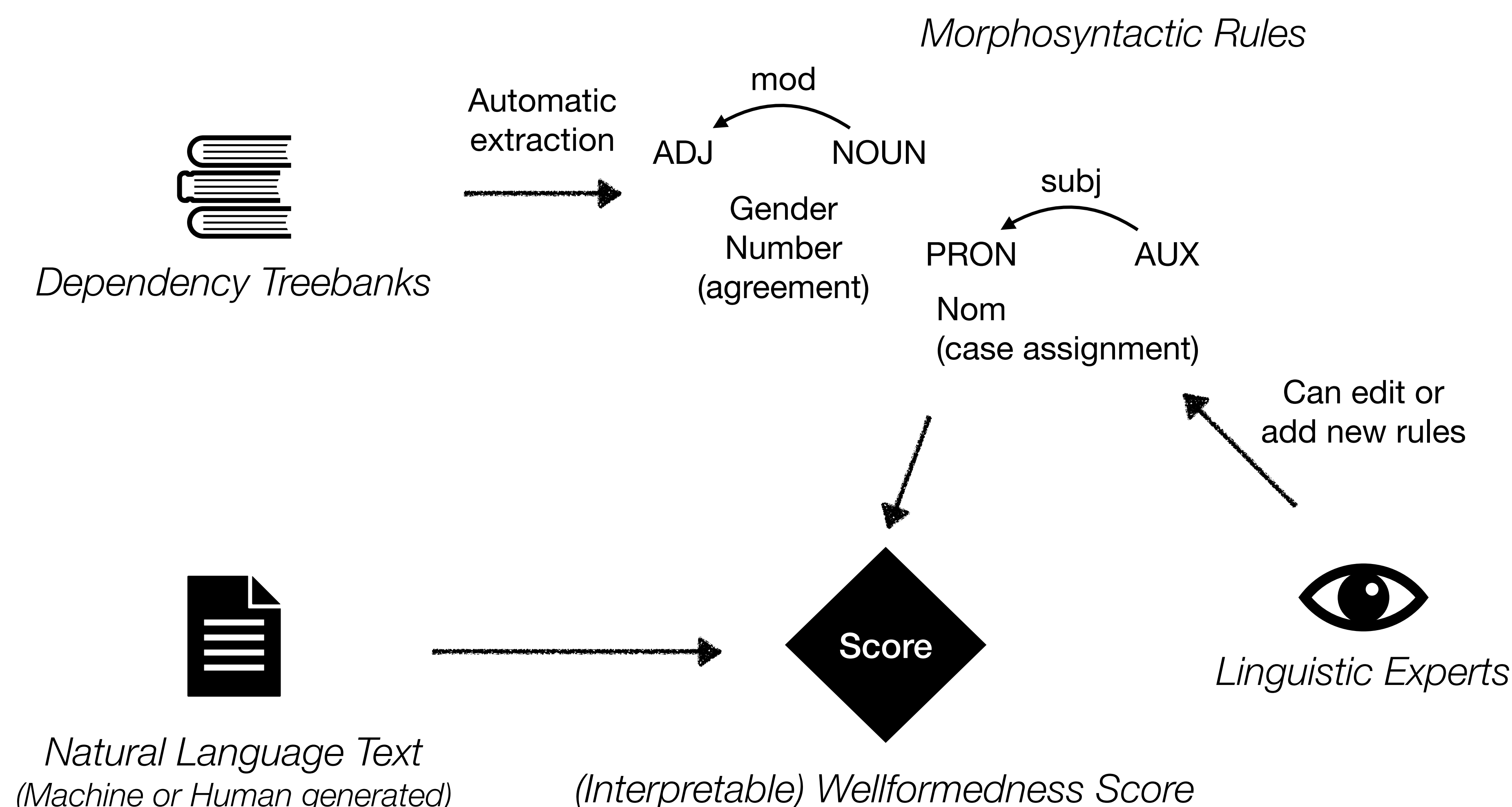


1. Grammatical Wellformedness

Analyzing wellformedness via the dependency structure and morphosyntactic rules.



2. L'AMBRE Overview



github.com/adithya7/lambre



5. Case Study: Neural Machine Translation

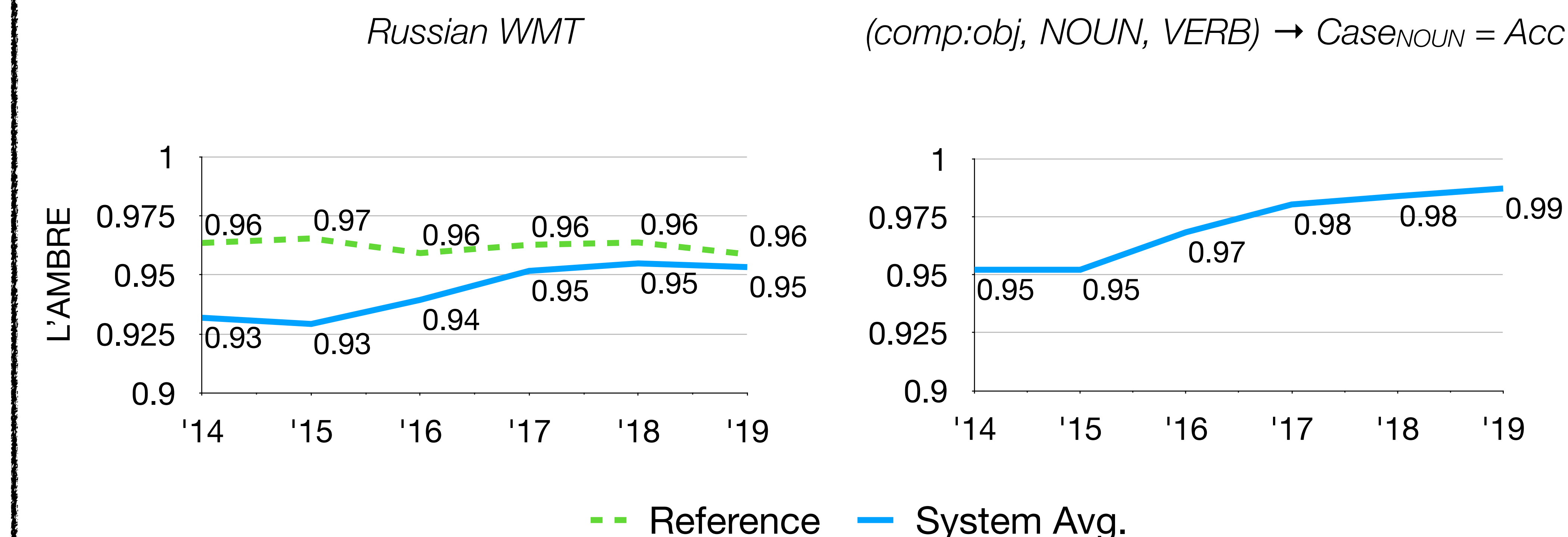
- We evaluate system outputs from WMT shared tasks
- L'AMBRE correlates well with human scores
- L'AMBRE correlates with perplexity, but more interpretable
- Diachronic analysis of systems and specific morphosyntactic rules

English → X	WMT '18	WMT '19
Czech	0.84	0.80
German	-0.06	0.16
Estonian	0.68	-
Finnish	0.86	0.85
Russian	0.86	0.57
Turkish	0.58	-

Pearson's r Correlations to human z-scores

Russian

(comp:obj, NOUN, VERB) → Case_{NOUN} = Acc

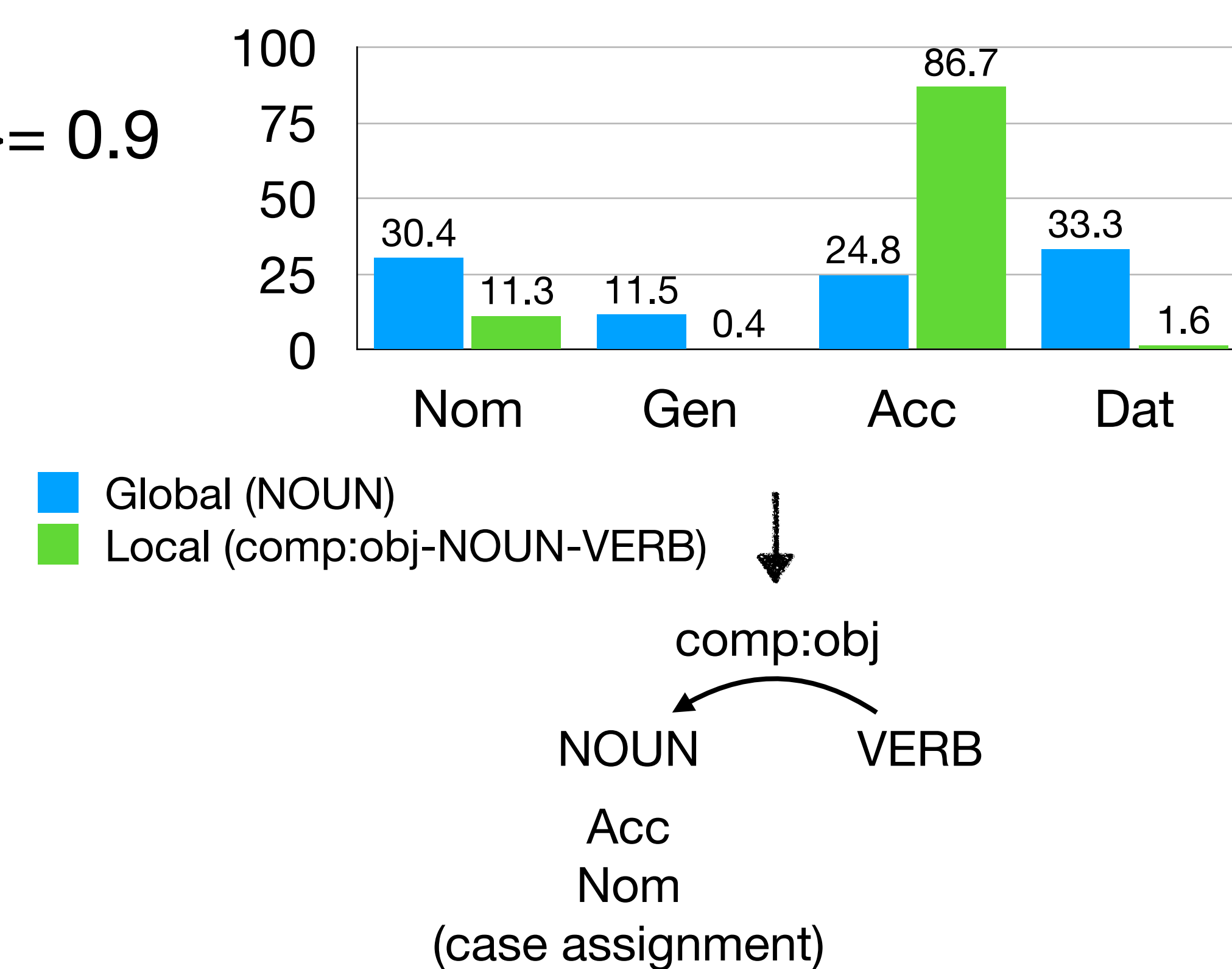
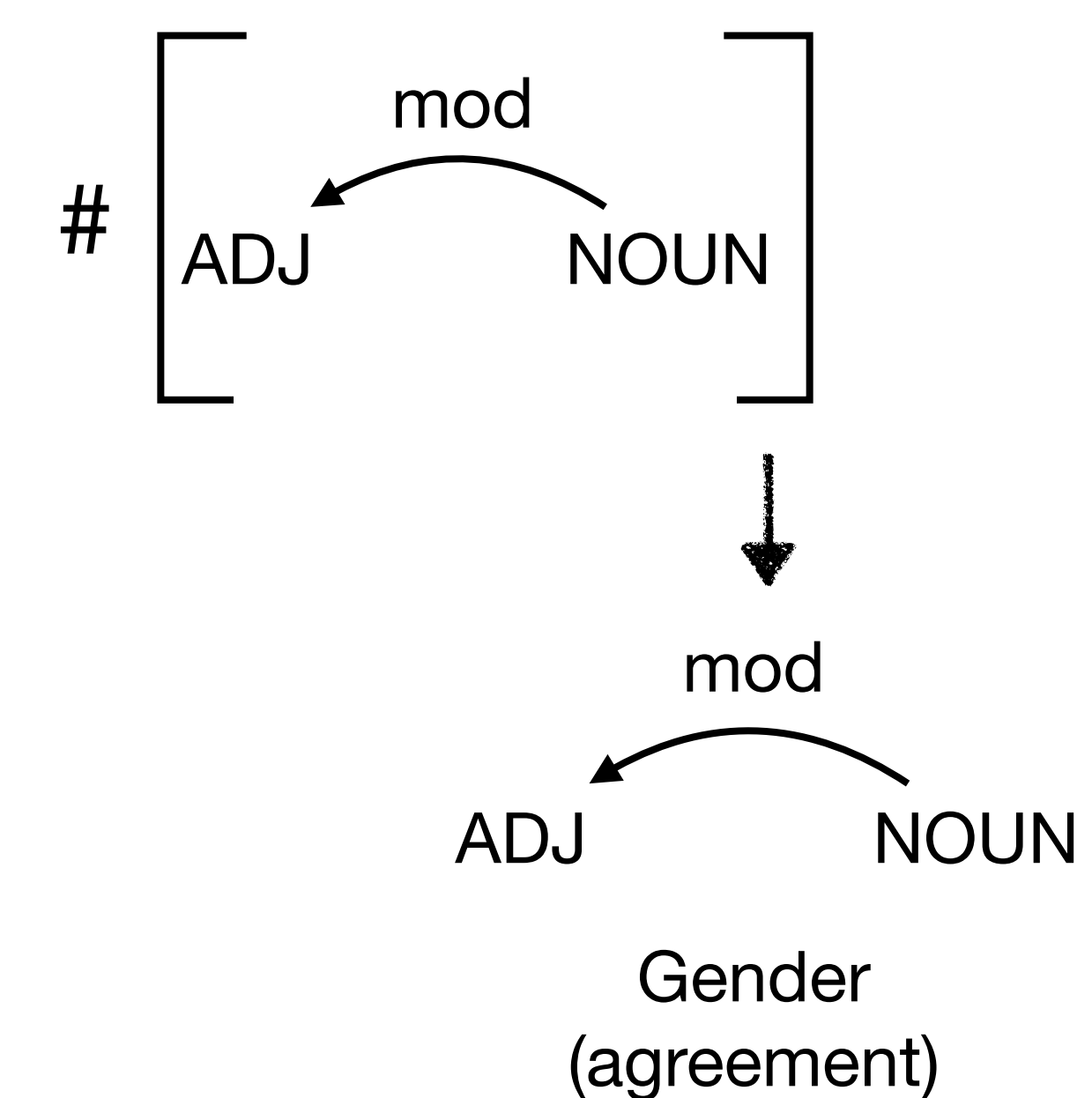


3. Rule Extraction

Illustration of extracting agreement and case assignment rules.

[Gender_{ADJ} = Gender_{NOUN}]

≥ 0.9



4. Building Robust Parsers

- Standard parsers and taggers are not robust to noisy text
- We utilize UniMorph to improve parser and tagger quality

